Cypher Anchor Smart Contract Security Audit

Comprehensive Architectural Review & Account Validation Invariants

PASSED & REMEDIATED

AUDITOR / ENGINE	DATE OF AUDIT	TARGET SCOPE	VERIFICATION COMMIT
Cypher Protocol Internal Security Committee	**November 2026**	**programs/cypher/src (crank.rs, position.rs, governance.rs, state.rs)**	`v1.0.0-release`

CRITICAL ISSUES	HIGH ISSUES	MEDIUM ISSUES	LOW ISSUES	STATUS
0	0	0	0	PASSED

1. Executive Summary

A rigorous, line-by-line security audit was performed on the core **Cypher** Anchor smart contract on Solana. The review focused on user collateral protection, cross-program invocation (CPI) safety, permissionless crank boundaries, PDA derivations, and mathematical precision.

2. Vulnerability Assessment & Remediation

Finding ID	Target / Vulnerability	Severity	Mitigation Status
CYPHER-SEC-001	Program Keypair committed in test fixtures	**CRITICAL**	**Permanently Deleted**
CYPHER-SEC-002	Shell command interpolation in freeze script	**HIGH**	**Hardened (spawnSync)**
CYPHER-SEC-007	Swap route account isolation missing vault_msol	**HIGH**	**Derived & Isolated**
CYPHER-SEC-008	Premature position close error verification	**MEDIUM**	**StrategyActive Guard**
CYPHER-SEC-010	Agent transaction preflight simulation missing	**MEDIUM**	**SimulateCheck Added**

3. Core On-Chain Invariants Verified

Security Invariant	Validation Logic	Result
Non-Custodial Vault PDAs	Funds reside exclusively in user PDAs; no admin key can withdraw collateral.	VERIFIED SECURE
Zero-Trust Cranks	Agent signer is strictly fee-payer; cannot alter mints, routes, or prices.	VERIFIED SECURE
CPI Route Account Isolation	User vaults derived and explicitly forbidden from swap route intermediary slots.	VERIFIED SECURE
Overflow Safety	All token math utilizes `checked_*` operations with u128 intermediate casts.	VERIFIED SECURE